

CUSTOMER CHURN ANALYSIS REPORT

Phase 1: Exploratory Data Analysis & Preprocessing

UK Banking Sector • Virtual Training Project • 2024

Tools: Microsoft Excel & Power BI

1,000

Total Customers

20.4%

Churn Rate

47%

Churn if Unresolved

52.2%

Resolution Rate

01 / OVERVIEW

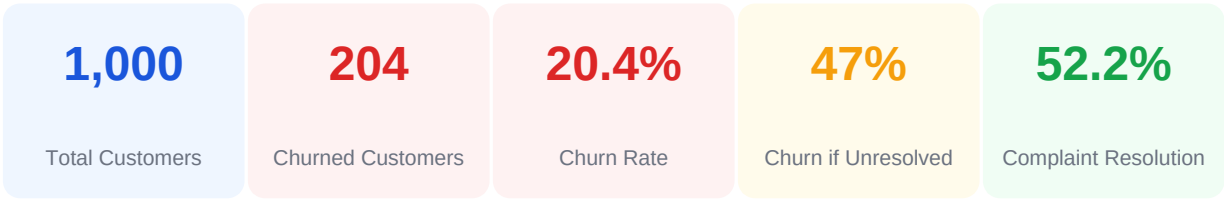
Project Background & Data Selection

A UK-based bank engaged a data analytics team to investigate a rising customer churn rate and identify the root causes behind customer departure. Phase 1 covers exploratory data analysis, key insight generation, and dataset preparation for predictive modelling.

Data Source	Rationale
Customer Demographics	Examines relationship between income levels and churn behaviour
Customer Service Interactions	Unresolved issues (48% of complaints) identified as primary departure driver
Online Activity / Service Usage	Evaluates whether digital channel choice (Mobile vs. Web) impacts loyalty

02 / KEY METRICS

At-a-Glance KPIs



03 / FINDINGS

EDA Key Insights

The following critical patterns were identified through the exploratory analysis of the 1,000-customer dataset across all three data sources:

#	Finding	Data Point
1	Overall churn rate exceeds banking benchmarks	204 / 1,000 customers churned (20.4%)
2	Unresolved complaints nearly double churn risk	Unresolved issue customers churn at 47% vs 20.4% average
3	Less than half of all complaints get resolved	Resolution rate stands at only 52.2%; 48% left unresolved
4	"Feedback" is the highest-failure complaint category	Feedback has the most unresolved cases of all interaction types
5	Low-income customers are the most at-risk segment	Low-income group has the highest churn volume across all brackets

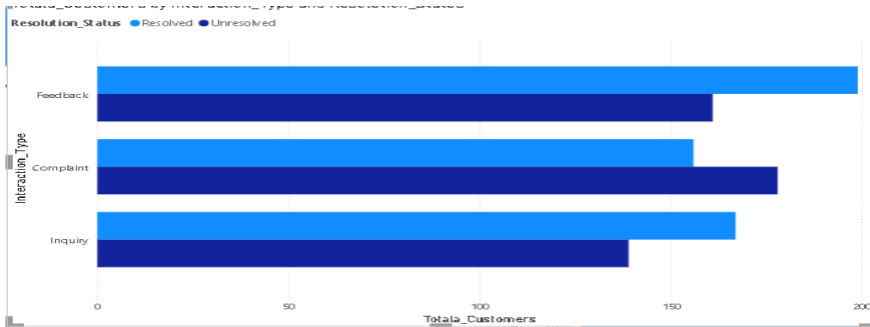
6	Online Banking is the top channel among churned users	38.73% of churned customers use Online Banking; Website 34.31%; Mobile 26.96%
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04 / VISUALIZATIONS

Supporting Charts

CHART 1

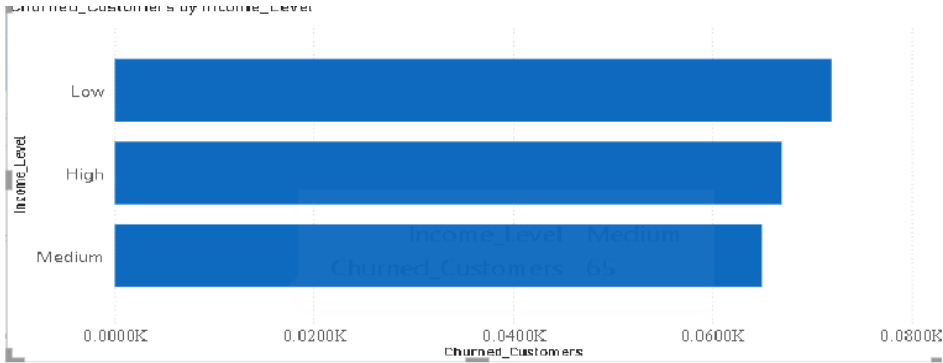
Complaints by Interaction Type and Resolution Status



Key Insight: Feedback is the interaction type with the highest absolute number of unresolved cases, making it the critical failure point in the bank's customer service pipeline.

CHART 2

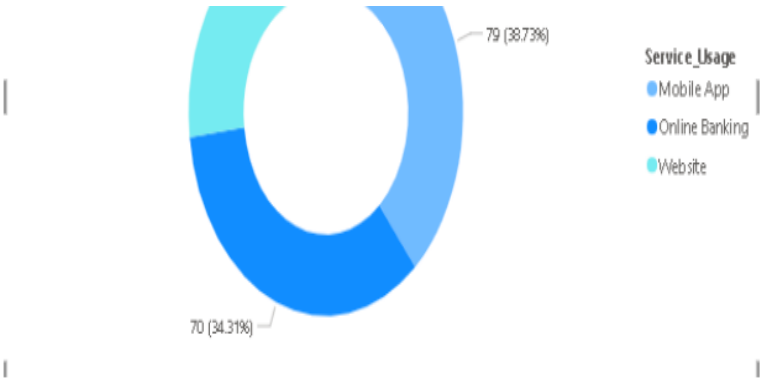
Churned Customers by Income Level



Key Insight: Low-income customers account for the highest volume of churned individuals, followed by High-income and then Medium-income segments.

CHART 3

Churned Customers by Service Channel



Key Insight: Online Banking is disproportionately represented among churned customers (38.73%), suggesting potential UX or service quality issues in the web platform.

05 / METHODOLOGY

Data Cleaning & Preprocessing

Before the dataset can be used for predictive modelling, four preprocessing steps were applied to ensure data integrity and model readiness:

- 1 Handling Missing Values**
Imputed missing entries in the income and service log fields to preserve the integrity of the full 1,000-customer dataset without introducing bias.
- 2 Categorical Encoding**
Applied One-Hot Encoding to Income_Level and Service_Usage variables, converting them to numerical format for compatibility with downstream machine learning algorithms.
- 3 Outlier Detection**
Inspected transaction volumes and interaction counts for statistical anomalies that could skew model training or distort segment-level analysis.
- 4 Normalization**
Scaled all numerical features to a consistent range, ensuring that high-magnitude variables (e.g., transaction amounts) do not dominate the model.

06 / RECOMMENDATIONS

Strategic Actions

Based on the EDA findings, three targeted interventions are recommended to address the 20.4% churn rate and its underlying drivers:

01

Fix the Feedback Resolution Pipeline

The Feedback category carries the highest unresolved volume. Assign dedicated agents to feedback cases and enforce a maximum 48-hour resolution SLA. Track resolution rates weekly as a KPI for the customer service team.

02

Launch a Low-Income Retention Programme

Design targeted offers for the low-income segment — fee waivers, simplified products, and proactive outreach when complaint activity spikes. This is the highest-volume churning group and the fastest win for retention.

03

Investigate the Online Banking Experience

With 38.73% of churned customers using Online Banking, there is likely UX friction or service gaps in the web platform. Conduct a UX audit and correlate platform issues with complaint types to identify root causes.

07 / CONCLUSION

Phase 1 Summary

The analysis clearly establishes that **Customer Service inefficiency** is the strongest predictor of churn. With a 47% churn rate among customers with unresolved complaints — compared to the 20.4% overall rate — resolving the feedback pipeline is the highest-impact intervention available. The Low-Income segment and Online Banking channel represent secondary priority areas. The dataset has been cleaned, encoded, and normalized and is now **ready for Phase 2 model development**.